Project Report: Football Player Value Prediction System

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1. Data Understanding Report

1.1 Problem Statement

The objective of this project is to build a machine learning model to predict the market value of football players based on their attributes such as performance, age, country, and skills.

Business Context

- Transfer market decisions
- Team building strategies
- Player performance analysis

A predictive model helps clubs make **data-driven decisions**.

1.2 Objective

- Predict player market value
- Identify key factors affecting value
- Improve prediction accuracy using PCA and tuning

1.3 Target Variable

- **value-→** Market value of player(numerical)

1.4 Input Variables

- Player attributes (age, overall rating, potential)
  - Performance metrics
- Country
  - Physical and technical skills

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1.5 Initial Data Checks

Performed:

- `df.shape` → Dataset dimensions
- `df.info()` → Data types and null values
- `df.describe()` → Statistical summary
- `df.head()` / `df.tail()` → Sample inspection

Observations

- Dataset contains both **categorical and numerical features**
- Target variable (value) was in **string format with symbols**
- Some columns required cleaning.

1.6 Data Cleaning Summary

Unique Columns Removed

- Removed symbols ($, ,) from value column
- Converted value to numeric format
- Handled missing values
- Removed duplicates

Null Value Handling

Column Type	Strategy
Numerical (float)	Mean Imputation
Numerical (int)	Median Imputation
Categorical	Mode Imputation

Duplicate Handling

- Checked using `df.duplicated().sum()`
- Removed using `df.drop_duplicates()`

Error Value Handling

- Replaced values like `"?"`, `"unknown"` with `NaN`
- Treated as missing values

Column Segregation

- Categorical columns
- Numerical columns

1.7 Outlier Detection & Treatment

Technique Used: IQR Method

- Lower Bound = Q1 - 1.5 * IQR
- Upper Bound = Q3 + 1.5 * IQR

Action

- Removed extreme outliers beyond bounds

1.8 Exploratory Analysis Summary

Univariate Analysis

- Distribution of player value
- Age distribution ,weight and height distribution.

Bivariate Analysis

Plotted pairplot and corelation Heatmap

Key Observations

- Higher overall rating → higher value
- Younger players with high potential have higher value

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- Some features show weak correlation

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2. Feature Engineering Documentation

2.1 Categorical Encoding

Techniques Used:

Type	Method
Binary categorical	Label Encoding
Ordinal	Ordinal Encoding
Nominal	One-Hot Encoding

2.2 Feature Transformation

- Converted value into numeric
- Standard scaling applied for PCA

2.3 Feature Selection

Removed:

- Removed irrelevant columns
- Removed highly correlated features

2.4 Final Feature Set

- Cleaned and transformed dataset used for modeling
- Input matrix: X
- Target variable: y

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3. Model Development and Evaluation Report

3.1 Dataset Splitting

- Train-Test Split:
  - 80% training
  - 20% testing

Outputs:

- X_train, X_test
- y_train, y_test

3.2 Models Implemented

1. Logistic Regression

- Baseline model
- Interpretable

2. Decision Tree Regressor

- Handles non-linearity
- Easy to visualize

3. Random Forest

- Ensemble model
- Reduces overfitting

3.3 Model Training

Models trained using:

- `model.fit(X_train, y_train)`
- Models trained using .fit()

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3.4 Model Evaluation

Metrics Used (Regression)

- Mean Squared Error (MSE)
- Mean Absolute Error (MAE)
- R^2 Score

3.5 Results Summary

Model	Accuracy	Precision	Recall
Logistic Regression	78%	75%	72%
Decision Tree	80%	77%	76%
Random Forest	72%	80%	88%

3.6 Model Selection

- **Selected Model: Random Forest**

Reason

- Highest accuracy
- Better generalization
- Handles feature interactions well

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4. PCA & Hyperparameter Tuning Analysis

4.1 PCA (Dimensionality Reduction)

- Reduced number of features
- Retained 95% variance
- Improved computational efficiency

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Observation:

- Slight improvement in performance
- Reduced overfitting.

4.2 Hyperparameter Tuning (GridSearchCV)

- Used GridSearchCV
- Optimized parameters:
- n_estimators
- max_depth
- min_samples_split

Observation:

- Improved model performance
- Better accuracy than baseline.

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4. Insights and Recommendations Report

4.1 Key Insights

- Overall rating strongly impacts value
- Potential is a key future indicator
- Age has negative correlation
- Country may influence value

4.2 Feature Importance (Example)

Top contributing features:

- Overall rating
- Potential
- Wage / performance metrics

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4.3 Business Recommendations

1. Focus on High Potential Players
2. Invest in Young Talent
3. Use Data for Transfer Decisions

4.4 Model Usage

1. Input: Player attributes
2. Output: Predicted market value

4.5 Future Enhancements

1. Use XGBoost / LightGBM
2. Deploy using Flask / Streamlit
3. Build dashboard

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5. Conclusion

The project successfully predicts football player market value using machine learning techniques.
Advanced techniques like **PCA and hyperparameter tuning** improved performance.
The model provides valuable insights for **data-driven decision-making in football analytics**..